Synoptic Question Playbook - Know Where to Start (Biology, Chemistry, Maths)

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1. Why Combined Questions Feel Impossible, and the START Method

Jody's teacher's second observation: when a question combines several topics, she does not know where to begin, and so she scores little or nothing. This is a skill problem, not a knowledge problem. A combined question is only several small questions wearing one coat. The fix is a fixed routine she runs every time, so that the first 60 seconds are never blank.

What actually goes wrong

What happens in the exam	Why it costs marks	The fix
She reads the question once and freezes	No working means no method marks, and no attempt means no credit for knowing the content	Always write something in the first minute: the START routine below gives her the first five lines
She tries to answer the whole question in one go	Long answers get muddled and miss mark points	Break the question into labelled mini-questions and answer each one separately
She does not notice which topics are involved	She uses the wrong idea, or only one of two needed ideas	Underline the topic keywords and write the topic name beside each
She writes a general answer that fits the topic	Marks are for the specific link the question asks for	Quote the data or context from the question in every sentence
She stops when she has one answer	Multi-mark questions need several linked points	Count the marks and match it with the number of distinct points

The START method (use it on every synoptic question)

Step	What to do	What to write down (so it is never blank)
S: Scan	Read the whole question and all data, graphs and tables. Circle the command word and the mark count.	The marks available, for example 6 marks means about 6 points or steps
T: Tag	Underline each key word and tag it with its topic (for example: enzyme = Topic 1, ATP = Topic 5).	A list of 2 to 4 topic tags down the margin
A: Arrange	Split the question into mini-questions in the order the examiner wants them (often the order of the data). Decide the order of the answer.	Numbered mini-questions: 1) what is the data showing, 2) why, 3) what does it mean for the whole system
R: Reply	Answer each mini-question in exam language, one point per sentence, using the question context.	A short paragraph or calculation per mini-question, with units and key terms
T: Tie together	Finish by linking the mini-answers into one conclusion that answers the exact question that was asked.	One closing sentence beginning Therefore, or Overall, using the words from the question

The 60-second rule

If Jody has no idea where to begin, she must still complete S, T and A on paper within one minute: marks, topic tags and mini-questions. Then she answers whichever mini-question is easiest first. Starting on an easy part nearly always unlocks the harder parts, and it secures the marks she knows.

Three linking phrases that gain marks

Phrase	Use it when	Example
This causes ... which leads to ...	Joining two topics in a chain	The enzyme is denatured, which causes fewer enzyme-substrate complexes, which leads to a lower rate of reaction

Phrase	Use it when	Example
This is because …	Giving the reason that the examiner wants	The rate is higher because the concentration of reactant molecules is greater, so there are more frequent collisions
Therefore … / Overall …	Closing with the conclusion	Therefore the data support the hypothesis that the drug reduces blood pressure

2. Question-Type Recognition: What Is It Really Testing, and What Is the First Move?

Use this table in the Scan step. Find the closest wording, then follow the first move.

Wording you see	What it is really testing	First move (write this down)
Explain how … and why … (6 marks)	A chain of cause and effect across two or more topics	Draw a small arrow chain in the margin: trigger, then step, then step, then result. Each arrow is a mark point.
Use the information provided and your knowledge …	Linking the data to a named biological or chemical idea	Quote one number or feature from the data, then name the idea that explains it
Evaluate / To what extent …	Weighing evidence for and against, with a judgement	Write two columns: supports and limits. Finish with a judgement sentence.
Discuss …	Several viewpoints or factors, then a conclusion	List 3 factors, give each a point and a reason, then conclude
Suggest why …	Applying knowledge to an unfamiliar situation	Ask: which topic is this really? Write the principle first, then apply it to the new context
Calculate … then hence deduce (Chemistry or Biology)	A multi-step calculation where each answer feeds the next	Write the chain: moles, ratio, then mass, concentration or percent. Label each line.

Wording you see	What it is really testing	First move (write this down)
Show that …	Reaching a given result with full working	Write the given answer at the top, then work forwards from the data to it
Find … then hence prove or find (Maths)	Combining two skills where part (a) is a tool for part (b)	Underline hence: it means use part (a). Write the part (a) result beside part (b).
Model the situation / use the model to …	Turning words into an equation, then interpreting	Define variables with units, write the equation, solve, then interpret the answer in the context of the question
Predict … and justify	A conclusion from a principle	State the principle (Le Chatelier, Hardy-Weinberg, chain rule), apply it, then state the prediction
Compare and contrast …	Similarities and differences using the same features	Choose 3 features and write one sentence for each feature covering both items
Interpret the graph / table …	Reading patterns and explaining them	Describe first (with figures from the graph), then explain using the correct concept

Golden rule for linking topics

When two topics appear in one question, ask: what is the bridge? Typical bridges:

Bridge idea	Biology example	Chemistry example	Maths example
Rate	Enzyme rate and temperature	Kinetics and activation energy	Rate of change as a derivative
Equilibrium or balance	Homeostasis feedback	Dynamic equilibrium and Kc	Stationary points
Quantity conserved	Mass flow in transport	Mole ratios	Equating coefficients
Structure and function	Protein shape and enzyme action	Bonding and properties	Form of function and graph shape

Bridge idea	Biology example	Chemistry example	Maths example
Energy	ATP and respiration	Enthalpy and bond energies	Work and kinetic energy in Mechanics

3. Biology (9BI0): Synoptic Questions Broken Down

Common topic combinations in the Biology papers

Combination	What the question looks like	Break it into mini-questions	Answer skeleton
Enzymes plus protein structure plus DNA	A mutation changes an enzyme and the rate of reaction drops	1) what changed in the DNA, 2) what changed in the amino acid sequence, 3) what changed in the tertiary structure and active site, 4) effect on enzyme-substrate complexes and rate	Substitution mutation, so different codon, so different amino acid; different R group interactions change the tertiary structure; the active site shape changes, so the substrate no longer fits, so fewer enzyme-substrate complexes, so the rate decreases
Respiration plus membranes plus transport	Why a poison that blocks the electron transport chain stops active transport	1) what the poison does, 2) effect on ATP production, 3) what active transport needs, 4) effect on uptake	Electron transport chain blocked, so less proton gradient and less ATP made by oxidative phosphorylation; active transport needs ATP to change carrier protein shape; so uptake against a concentration gradient stops

Combination	What the question looks like	Break it into mini-questions	Answer skeleton
Photosynthesis plus limiting factors plus data	A graph of rate against light intensity then levelling off	1) describe the trend with figures, 2) name the limiting factor in each region, 3) explain using the stage affected	At low intensity light is limiting because the light-dependent reaction makes less ATP and reduced NADP; the plateau means another factor such as carbon dioxide concentration or temperature is limiting the light-independent reaction
Genetics plus meiosis plus evolution	Explain how a new allele could become common in a population	1) source of the allele, 2) how it is passed on, 3) selection pressure, 4) change in allele frequency	Mutation creates the allele; passed on in gametes through meiosis and fertilisation; if it gives a selective advantage individuals survive and reproduce more, so the allele frequency increases over generations
Immunity plus cell cycle plus DNA	Why a vaccine gives memory and how B cells divide	1) antigen recognition, 2) clonal selection and expansion by mitosis, 3) memory cells and secondary response	Antigen presented, specific B cell with complementary receptor selected, divides by mitosis to form plasma cells and memory cells; on re-infection memory cells respond faster and more strongly

Combination	What the question looks like	Break it into mini-questions	Answer skeleton
Cardiovascular plus gas exchange plus surface area ratio	Why a larger organism needs a transport system	1) surface area to volume ratio, 2) diffusion distance, 3) metabolic demand, 4) the system that solves it	Larger organism has smaller SA:V ratio and longer diffusion distance, so diffusion alone is too slow for the metabolic demand; a mass transport system maintains steep concentration gradients
Homeostasis plus hormones plus negative feedback	Blood glucose control after a meal	1) stimulus, 2) receptor, 3) coordinator, 4) effector, 5) response and negative feedback	Rise in glucose detected by beta cells; insulin released; liver and muscle take up glucose, convert to glycogen via glycogenesis; glucose falls and this reduces insulin secretion, which is negative feedback
Ecology plus statistics plus sampling	Compare two sites and decide if the difference is significant	1) sampling method, 2) description, 3) choice of statistical test, 4) conclusion	Random quadrats using random coordinates; compare means; Spearman or t-test as appropriate; compare the test value with the critical value at p = 0.05; conclude with reference to the null hypothesis

Worked example (6 marks): A mutation and the rate of respiration

Question: A mutation in the gene for a respiratory enzyme in mitochondria leads to a person having less ATP available for muscle contraction. Explain how this could happen. (6 marks)

S (Scan): 6 marks, command word explain, topics: genes, protein structure, enzymes, respiration, muscle.

T (Tag): mutation = Topic 4 (genes), enzyme = Topic 1, respiration = Topic 5, muscle = Topic 5 or 6.

A (Arrange): 1) DNA change, 2) protein change, 3) enzyme change, 4) respiration change, 5) ATP and muscle.

R (Reply): A mutation changes the base sequence of the gene, so the sequence of codons changes and a different amino acid may be coded for. This changes the primary structure and therefore the hydrogen bonds and ionic bonds that form the tertiary structure. The shape of the active site changes, so the substrate is no longer complementary and fewer enzyme-substrate complexes form. The rate of respiration decreases, so less ATP is made by oxidative phosphorylation. Muscle contraction needs ATP for myosin head movement, so less ATP means weaker contraction.

T (Tie together): Therefore the mutation reduces the enzyme activity, which reduces ATP production and muscle contraction.

Eight habits that gain marks on Biology synoptic questions

Habit	Example
Quote the data	At 30 degrees Celsius the rate is 12 arbitrary units, compared with 4 at 20 degrees Celsius
Use the context word	The enzyme in this question, not the enzyme
One idea per sentence	Do not join three ideas with and
Use the chain words	so, therefore, which leads to
Name the structure and the function	The cristae provide a large surface area for the electron transport chain
Use comparative language	higher, lower, more, fewer, greater rate of
Give the number of marks as points	A 6-mark answer should have at least 6 distinct points
End with the conclusion	Therefore …

4. Chemistry (9CH0): Synoptic Questions Broken Down

Common combinations in the Chemistry papers

Combination	What the question looks like	Break it into mini-questions	Answer skeleton
Titration plus moles plus percentage yield or purity	A solid is dissolved and titrated, find the percentage purity	1) moles of titrant, 2) ratio from the balanced equation, 3) moles in the whole sample, 4) mass, 5) percent purity	n = c x V (in dm3); use the mole ratio; scale up for the full volume; mass = n x Mr; purity = mass of pure over mass of sample x 100
Equilibrium plus kinetics plus energetics	Effect of temperature on yield and rate for an exothermic reaction	1) position of equilibrium, 2) rate, 3) compromise	Higher temperature moves the equilibrium to the endothermic side, so lower yield (Le Chatelier); but rate increases because more particles have energy greater than Ea; so a compromise temperature is used
Born-Haber cycle plus lattice energy plus bonding	Why one lattice enthalpy is more exothermic	1) ionic radius, 2) ionic charge, 3) attraction, 4) comparison	Smaller radius or higher charge gives stronger electrostatic attraction, so more exothermic lattice enthalpy
Organic synthesis plus mechanisms plus spectra	Identify an unknown from IR, NMR and mass spectrum, then name the mechanism	1) molecular ion gives Mr, 2) IR peaks give functional groups, 3) NMR gives environments, 4) assemble the structure, 5) mechanism	Use M+ for Mr; peak at 1700 cm-1 means C=O; count environments and splitting; propose the structure; check it fits all data; draw the mechanism with curly arrows

Combination	What the question looks like	Break it into mini-questions	Answer skeleton
Kinetics plus rate equation plus mechanism	Deduce order from data and propose the rate-determining step	1) order for each reactant, 2) rate equation, 3) k and units, 4) mechanism	Compare how rate changes when concentration changes; write rate = k[A][B]^2; find k with units; species in the rate equation are in the rate-determining step
Acid-base plus buffers plus pH	Find the pH of a mixture or a buffer	1) moles of acid and base, 2) decide excess, 3) use the right pH expression	Strong acid with strong base: excess H+ gives pH; weak acid with its salt: use Ka and the ratio for a buffer
Electrochemistry plus redox plus thermodynamics	Predict whether a reaction is feasible from E values	1) half-equations, 2) E cell = E reduction minus E oxidation, 3) sign, 4) caveat on kinetics	If E cell is positive the reaction is feasible, but it may be slow because of high activation energy
Transition metals plus complex ions plus colour	Explain colour and ligand exchange	1) d-orbital splitting, 2) energy gap, 3) frequency absorbed, 4) colour seen	Ligands split the d orbitals; electrons are promoted; the energy gap matches the frequency of light absorbed; the complementary colour is seen

The multi-step calculation chain (use this for all calculations)

Step	Write	Common mark lost
1	The balanced equation	Not balanced
2	Moles of the known substance with units	Using cm3 not dm3
3	The mole ratio from the equation	Ratio left out

Step	Write	Common mark lost
4	Moles of the required substance	Wrong scale factor
5	Convert to the quantity requested (mass, concentration, volume)	No units
6	Round to the number of significant figures in the data	Rounding too early

Worked example (6 marks): Explain the conditions for the Haber process

Question: The Haber process is carried out at 450 degrees Celsius and 200 atm with an iron catalyst. Explain why these conditions are used. (6 marks)

S: 6 marks, command word explain, three conditions: temperature, pressure, catalyst.

T: temperature = equilibrium (Topic 10) and kinetics (Topic 9), pressure = equilibrium, catalyst = kinetics and energetics.

A: 1) the equilibrium and the enthalpy sign, 2) temperature trade-off, 3) pressure trade-off, 4) catalyst.

R: The forward reaction is exothermic, so a lower temperature would increase the equilibrium yield according to Le Chatelier. However, a lower temperature gives a slower rate because fewer particles have energy greater than the activation energy, so 450 degrees Celsius is a compromise between yield and rate. There are fewer moles of gas on the right (2 against 4), so a higher pressure moves the position of equilibrium to the right and gives a higher yield; high pressure also increases the rate by increasing the frequency of collisions. 200 atm is chosen because higher pressures are expensive and need stronger equipment. The iron catalyst provides an alternative route with a lower activation energy, so the rate increases without changing the position of equilibrium or the yield.

T: Therefore the conditions are a compromise between yield, rate and cost.

Eight habits that gain marks on Chemistry synoptic questions

Habit	Example
Write the balanced equation first	N2 + 3H2 forms 2NH3
Name the principle	By Le Chatelier's principle
Distinguish rate from yield	Rate is how fast; yield is how much

Habit	Example
Use particle language	More particles with energy greater than or equal to the activation energy
State units on every line	mol dm-3, kJ mol-1, J K-1 mol-1
Show the sign of the enthalpy change	Exothermic, delta H negative
Draw the mechanism with curly arrows to the correct atoms	Arrow from lone pair or bond to the atom
Check the final answer is realistic	Percent yield cannot exceed 100

5. Maths (9MA0): Synoptic Questions Broken Down

In Maths a synoptic question usually hides several topics behind one story. The skill is to name the topics, then use the answer of each step as the input of the next.

Common combinations in the Maths papers

Combination	What the question looks like	Break it into mini-questions	Answer skeleton
Differentiation plus trigonometry plus modelling	A tide height model h = a + R sin(bt + c), find the time of maximum rate of rise	1) differentiate to get dh/dt, 2) set the derivative equal to a value or find where it is maximum, 3) solve the trig equation in range, 4) interpret in context	dh/dt = Rb cos(bt + c); maximum when cos = 1; solve bt + c = 0 plus 360n; convert to a time and state it with units
Logs plus exponentials plus modelling	Data follow y = a x^n or y = a b^x, find a and n	1) take logs of both sides, 2) compare with y = mx + c, 3) read gradient and intercept, 4) convert back	log y = log a + n log x, so gradient n and intercept log a; a = 10 to the power intercept

Combination	What the question looks like	Break it into mini-questions	Answer skeleton
Series plus logs	The first year the total exceeds a target	1) identify the series type, 2) write the sum formula, 3) form the inequality, 4) use logs to solve for n, 5) interpret as a whole number of years	S_n greater than target; take logs; remember dividing by a negative log reverses the inequality; round up to the next integer
Integration plus area plus parametrics	Area under a parametric curve	1) dx/dt, 2) convert the limits to t, 3) set up the integral of y dx/dt, 4) integrate (often trig identity), 5) evaluate	Area = integral of y (dx/dt) dt; limits in t; use double angle formula to integrate sin squared
Differentiation plus integration plus differential equations	Rate problem leading to a separable differential equation	1) form the equation from the words, 2) separate variables, 3) integrate with constant, 4) use initial condition, 5) answer in context	dV/dt = -kV; integral dV/V = integral -k dt; ln V = -kt + c; V = A e to the power minus kt
Proof plus algebra	Prove that the sum or product of numbers has a property	1) write general forms, 2) expand and simplify, 3) factorise to show the property, 4) conclude in words	Let the numbers be n and n + 1; expand; take out a common factor; this is a multiple of 2 so it is even
Vectors plus trig plus geometry	Angle between two lines or position of a point	1) write vectors, 2) use dot product or ratio, 3) find angle or point, 4) check in context	cos theta = a dot b over magnitude a times magnitude b
Statistics plus probability plus binomial	Hypothesis test following a probability model	1) define the model, 2) state hypotheses in context, 3) calculate p-value or critical region, 4) conclude	X ~ B(n, p); H0 and H1; compare with significance level; conclusion in context

Combination	What the question looks like	Break it into mini-questions	Answer skeleton
Mechanics plus calculus plus trig	Projectile or variable force with resolving	1) resolve into components, 2) apply suvat or integrate, 3) combine components, 4) answer in context	Resolve horizontally and vertically; use time as the common link between components

The topic-ladder method

When a Maths question gives no clue where to begin, climb a ladder of four questions.

Rung	Ask	Write
1	What is given?	List every number, function and condition
2	What is asked?	Write the target in symbols, for example find t when dh/dt is greatest
3	What tool bridges the two?	Name the topic: differentiate, take logs, form a quadratic in a single function
4	What is the first line of working?	Write the formula or the equation, even if you cannot finish yet

Worked example (8 marks): Population model

Question: A population is modelled by P = 200 e^(0.05t). (a) Find the initial population. (b) Find the rate of increase when t = 10. (c) Find, to 1 decimal place, when the population first exceeds 1000. (d) Comment on the model.

S: four parts; topics are exponential functions, differentiation, logarithms and interpretation.

T and A: (a) substitute t = 0; (b) differentiate then substitute; (c) form inequality then take natural logs; (d) limitation.

R: (a) At t = 0, P = 200 e^0 = 200. (b) dP/dt = 200 x 0.05 e^(0.05t) = 10 e^(0.05t); at t = 10, dP/dt = 10 e^0.5 = 16.5 (3 s.f.), so the population is increasing at 16.5 per unit time. (c) 200 e^(0.05t) greater than 1000, so e^(0.05t) greater than 5; take natural logs: 0.05t greater

than ln 5, t greater than 32.2 (1 d.p.), so the population first exceeds 1000 after 32.2 time units. (d) The model predicts unlimited exponential growth, which is unrealistic because resources are limited.

T: Each part used the previous formula, and all answers are stated in context with units.

Eight habits that gain marks on Maths synoptic questions

Habit	Example
Label every part	(a), (b), (c) and write the topic in the margin
Use the previous part	Hence, use the result of part (a)
Write the formula before the substitution	A = 1/2 r squared theta, then substitute
Keep exact values until the end	Leave as ln 5 or root 3 until the final line
Check the range	Is the answer in the interval asked
Interpret the result	The population first exceeds 1000 after 32.2 years
Test one value	Substitute back to check the equation
Attempt every part	Method marks are available even without the final answer

6. Cross-Subject Skills: Maths Inside Science, Graphs and Data

Maths skills that appear inside Biology and Chemistry questions

Skill	Where it appears	Method to write down
Percentage change	Growth, yield, mass loss	(final minus initial) over initial x 100, with the sign shown
Rate from a graph	Enzymes, kinetics	Draw a tangent at the point, then rate = change in y over change in x, with units
Magnification	Microscopy	Magnification = image size over actual size; convert units first (mm to micrometres)

Skill	Where it appears	Method to write down
Standard form and units	Chemistry calculations	Convert to dm3, g, mol first; keep standard form and the right number of significant figures
Ratios and proportions	Genetics, moles	Simplify the ratio, then apply it to the actual numbers
Log scales	pH, kinetics, population growth	pH = minus log10 of H+ concentration; a change of 1 pH unit is a ten-fold change
Rearranging equations	Rate equations, Kc	Change the subject in steps, keeping the units consistent
Statistical tests	Ecology, genetics	State null hypothesis, choose test, calculate, compare with the critical value, conclude in context
Standard deviation and error bars	Data questions	Overlapping error bars suggest the difference may not be significant

Reading a graph or table in three layers

Layer	Question	Example sentence
1 Describe	What does the data show? Use figures.	The rate increases from 2 to 10 units between 10 and 40 degrees Celsius, then falls to 1 unit at 60 degrees Celsius
2 Explain	Why? Use the correct concept.	The rate rises because molecules have more kinetic energy, so more successful collisions; it falls because the enzyme is denatured
3 Conclude	What does it mean for the question asked?	Therefore the optimum temperature is about 40 degrees Celsius

Data question checklist

Check	Question
Axes	What are the variable and units on each axis?

Check	Question
Trend	Is it increasing, decreasing, a peak, a plateau?
Numbers	Have I quoted two values from the graph with units?
Anomaly	Is there an outlier and can I suggest a reason?
Limits	Is the sample size small, is there a control, is there repeatability?
Link	Which topic explains this pattern?

7. Daily Practice Routine, Checklist and Tracker

The 20-minute daily drill (three questions a day)

1. Take one synoptic question (from the mock papers or the flashcards). Set a 3-minute timer.
2. In those 3 minutes complete only S, T and A on paper: marks, topic tags and mini-questions. Do not answer yet.
3. Compare with the model decomposition. Mark whether the tags and mini-questions match.
4. Answer the question in full within the time for the marks (about 1.5 minutes per mark in Biology and Chemistry, 1 minute per mark in Maths).
5. Mark the answer against the model. Write one sentence: which link or step did I miss?

Weekly plan

Day	Focus
Monday	Biology: two linked-topic questions and one data question
Tuesday	Chemistry: one multi-step calculation and one explain question
Wednesday	Maths: two combined-topic questions using the topic ladder
Thursday	Weakest subject from the week, timed

Day	Focus
Friday	One full 6 to 9 mark answer in each subject, written without notes
Weekend	Review the mistake log and redo two questions

Self-check before moving on

Check	Question
Started in 60 seconds	Did I write marks, tags and mini-questions before answering?
Topics named	Did I identify every topic involved?
Bridge found	Did I link the topics with a cause-and-effect chain?
Context used	Did I quote data or the context from the question?
Points counted	Did I give at least as many distinct points as marks?
Conclusion	Did I finish with a sentence that answers the exact question?

Progress tracker

Date	Subject and question	Time to first line (seconds)	Marks	Link I missed

Note for the tutor

Target for the next four weeks: first line written within 60 seconds on every combined question, then 70 percent of available marks on 6 to 9 mark synoptic answers. The worked examples are models; after the next mock paper, add Jody's real mistakes to the tracker so the drills target what actually costs her marks. Pair this playbook with the Exam Language Banks: this one teaches how to start and structure, the language banks teach how to phrase.